

Limenet

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Limenet

Company or organization location (we welcome applicants from anywhere in the world)

Lecco, Lombardy, Italy

Name(s) of primary point(s) of contact for this application

Stefano Cappello

Brief company or organization description <20 words

Low-cost CO₂-free lime (CaO) to remove CO₂ through the atmosphere using a novel design for electric calcination.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Mineral OAE has a great deal of potential for large scale carbon removal. However, to make this pathway scalable and efficient, there needs to be a scalable feedstock of low or no emissions OAE minerals like calcium oxide and magnesium oxide.

Limenet is working on a state-of-the-art technology that generates low-cost CO₂-free calcium oxide (CaO) using **limestone**, **renewable electricity** and **seawater**. The CO₂-free lime is then used to make a measurable and permanent carbon removal.

To produce it, Limenet starts with a novel design electric calciner with an integrated CO₂ capture system that heat up limestone using a proprietary indirect vertical electric kiln. Here limestone is reaching calcination temperatures with electric resistance and is transformed into lime and CO₂.

To produce CO₂-free lime, the CO₂ produced during calcination process is stored using Limenet technology: an **Ocean Alkalinity Enhancement** approach (**pH equilibrated**). Calcination CO₂ and 50% of the lime are conveyed into a mixer where, inside a flow of seawater, they are transformed into a solution of calcium bicarbonate reaching a pH-equilibrated solution that is injected into seawater with the same background sea pH.

The remaining 50% of lime is CO₂-free lime.

To make the carbon removal, the 50% of the CO₂-free lime could be used in different ways.

- 1) It can store CO₂ coming from biogenic sources, DAC facilities, or gasification units processing biogenic materials. This is done using Limenet storage technology.

Or it can be used to capture CO₂ from the atmosphere if the CO₂-free lime is exposed to the atmosphere. This can be done in two ways:

- 2) CO₂-free lime sprayed on a land surface, and it carbonates back into limestone (Passive Direct Air Capture through mineralization).
- 3) CO₂-free lime is used to make mineral OAE (Ocean liming) thus transformed into bicarbonates.

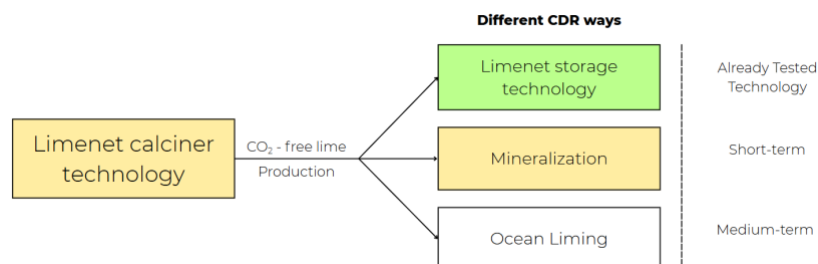


Figure 1 Way of use of CO₂ - free lime

Storage:

Limenet's technology permanently stores, for thousands of years (> 10,000 years)^{2,3}, all the CO₂, both produced by the process (from Limestone), from external sources or the atmospheric CO₂, as calcium bicarbonates into seawater using a unique and completely innovative marine CDR (Mineral OAE) way of producing a pH Equilibrated Alkaline Solution.

² Renforth, P., & Henderson, G. (2017). Assessing ocean alkalinity for carbon sequestration. *Reviews of Geophysics*, 55(3), 636–674. <https://doi.org/10.1002/2016RG000533>

³ ISOMETRIC, Dissolved Inorganic Carbon Storage in Oceans.

Technology modules:

The technology consists of 2 different modules:

- An **electric indirect kiln (Calciner)**, powered by renewable electricity, converts Limestone (CaCO_3) into lime (CaO) and fossil CO_2 . The fossil CO_2 is stored in the oceans in the form of a pH equilibrated calcium bicarbonates solution using 50% of the lime previously produced. With that system it is possible to have the other 50% of lime (CO_2 -free lime) produced with no process CO_2 emissions. This 50% of CO_2 -free lime is used to make CDR.
- A **Mixer** generates calcium bicarbonates reacting seawater, fossil CO_2 (produced by the calcination of limestone) or atmospheric CO_2 (captured from the atmosphere through carbonation) and the CO_2 -free lime produced by the calciner. The output of the mixer is a transparent aqueous solution of calcium bicarbonates with a pH equilibrated to that of seawater⁴. Once injected into the ocean, the bicarbonates solution is considered a stable form of stored CO_2 that has a co-benefit that can counteract the acidification of seas and oceans thanks to its buffering effect.

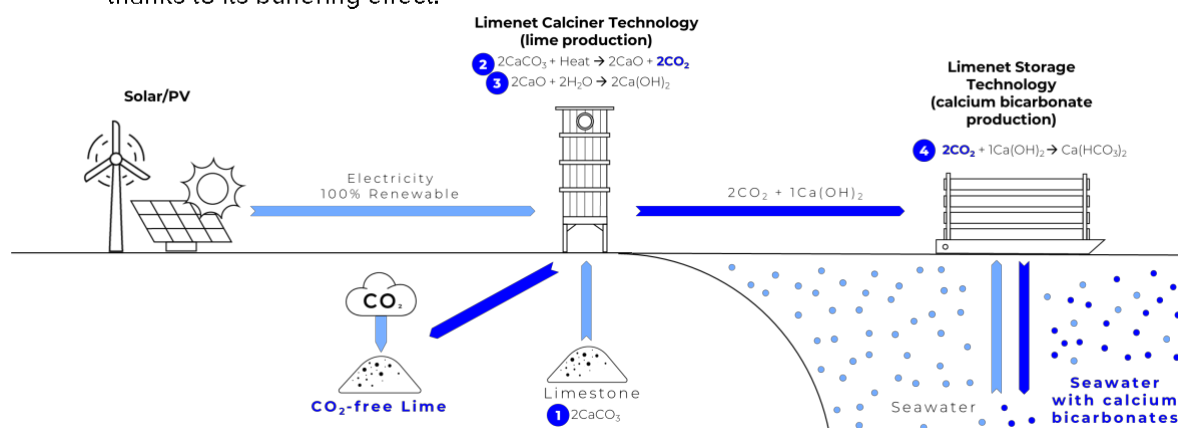


Figure 2 - Limenet's process

⁴ Varliero, S., Jamali Alamooti, S., Campo, F. P., Cappello, G., Cappello, S., Caserini, S., Comazzi, F., Macchi, P., & Raos, G. (2024). Assessing the limit of CO_2 storage in seawater as bicarbonate-enriched solutions. *Molecules*, 29(17), 4069. <https://doi.org/10.3390/molecules29174069>

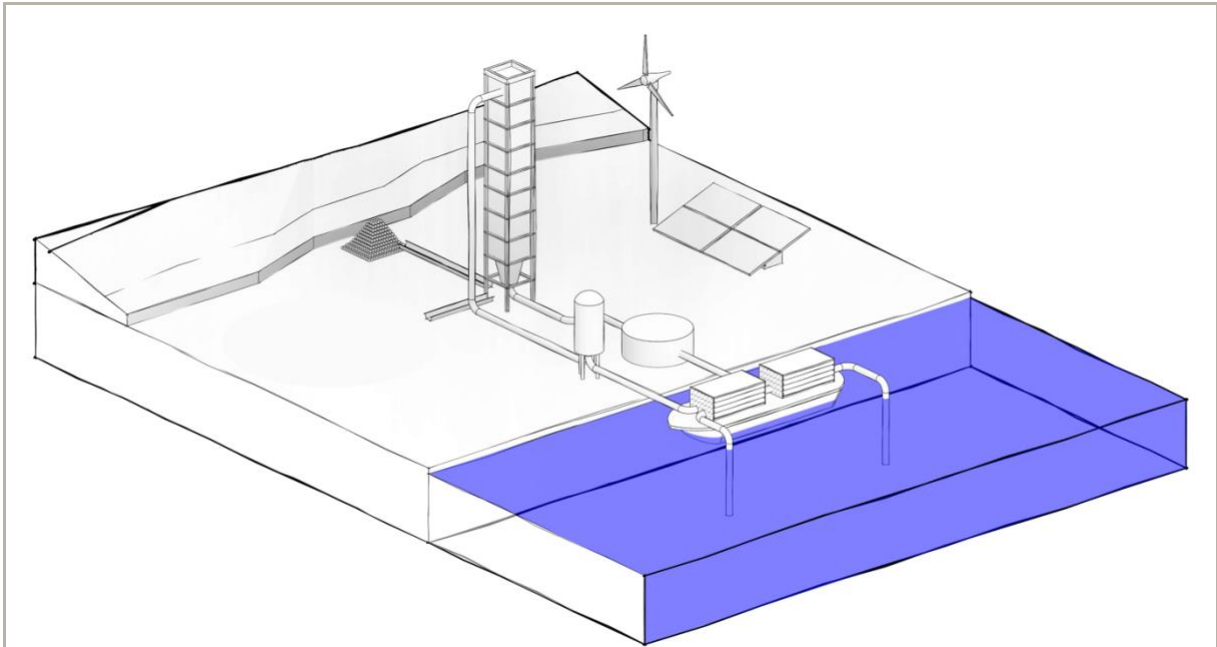


Figure 3 - FOAK TRL9 plant 3D render

Vision of Limenet project:

Limenet in its industrial strategy envisions a future with a technology able to produce CO₂-free lime to make Carbon Dioxide Removal. In particular Ocean Liming pathway. However, to achieve this, more research should be done to decrease the VCL. For this reason, we believe the flexibility to use the same CO₂-free lime in different CDR pathway could be a derisking for Frontier, investor, policy maker and for this early phase of the market.

Characteristics of Limenet project:

Below is a list of the main features of the Limenet process that we believe qualify it as a “great project”:

- **Co-Benefits:** injecting alkalinity into the ocean will counteract ocean acidification and preserve marine biodiversity.
- **Measurement based MRV:**
 - o all the parameters to assess the quantity and the quality of the CO₂ stored into seawaters are measured and verified.
 - o backed by more than 15 scientific papers, Limenet is using just science-based methodology for accounting from LCA to experimental studies.
- **Integration with existing infrastructure:** one of Limenet’s key objectives is to install its plants within existing industrial port areas to avoid the costs associated with building dedicated infrastructure. The flexibility and modularity of Limenet’s technology allow for easy installation as it does not require major infrastructural work.
- **Robust feedstock sourcing:** Although the availability of limestone is about 5,000 gigatons (Gt) within 10 km of the coast worldwide⁵, Limenet will prioritize the use of limestone fines already stocked in quarries because traditional liming technologies are unable to process fine-grained material. In this way, Limenet intends to valorize and give new life to waste that would otherwise remain stored in quarries and avoid taking material away from other sectors in which calcium carbonate is used.

⁵ Caserini et al., 2022. The Availability of Limestone and Other Raw Materials for Ocean Alkalinity Enhancement. *Global Biogeochemical Cycles*, 36, 5, e2021GB007246. <https://doi.org/10.1029/2021GB007246>

- **No need to manage byproducts or contaminants:** the Limenet process does not generate any byproducts or contaminants that require disposal or treatment.
- **No need to build new supply chains:** the supply chain for calcium carbonate is well-established and globally available and widely used in multiple industries (construction, paper, plastics, agriculture). This ensures reliable sourcing, cost efficiency, and scalability for Limenet's operations.

Our Strengths:

- **All in one solution:** Limenet is one of the few carbon removal companies that integrates CO₂ capture from the atmosphere and storage within into a single process. Unlike most other companies, which rely on external partners for either capturing or storing CO₂, Limenet provides an end-to-end solution. By integrating both steps, the system eliminates the need for separate CO₂ transportation, or geological injection infrastructure.
- **Flexible technology:** Limenet technology enables the development of three different approaches (Direct Air Capture-DAC, Ocean Alkalinity Enhancement-OAE, Ocean Storage into bicarbonates-OS) for removing CO₂ from the atmosphere that will be deployed depending on the resources and the energy available in the region of deployment:
 - o **Carbonation lime (mineralization):** CO₂-free lime is mineralized through carbonation of Lime (DAC) by spreading it on the ground in thin layers. Lime removes CO₂ from the atmosphere. The remaining half of lime is still used for OS.
 - o **Liming:** CO₂-free lime is used for ocean liming (OAE). The addition of lime to the ocean spreading by means of barge increases alkalinity and removes CO₂ from the atmosphere. The remaining half of lime is still used for OS.
 - o **Removing CO₂ coming from biogenic or other sources such as the upgrading of biogas (OS):** CO₂-free lime is used to store biogenic CO₂ coming from the process of upgrading biogas to biomethane.
- **Innovative kiln:** the Limenet kiln is an innovative indirect-electric kiln, designed to separate the combustion and calcination chamber, thus avoiding contamination from fuels. Its flexible design allows the use of both renewable energy and biomass-derived syngas, with low thermal inertia during start-up and shutdown, and it can be relocated as needed thanks to its modularity. It is capable of processing small-sized limestone or even fine powder, making possible the use of material that is typically considered waste and generate CO₂-free CaO. All of this comes with a lower CAPEX compared to traditional technologies.
- **Lower CapEx cost compared to "traditional" DAC solutions:** Limenet uses of the shelf's components with no exotic materials or components. For that reason, according to Limenet TEA compared to others competitor analyzed in the paper: Considering technology characteristics to project future costs of direct air capture⁶ and Frontiers applicants, Limenet has lower capex on passive DAC system that results in a lower levelized cost (with same assumption).
- **Injection of a pH Equilibrated Alkaline Solution of calcium bicarbonate into the sea:**
 - o Injecting the alkaline solution with the same seawater pH decreases the risk of abiotic and biotic carbonates precipitation. In fact, it avoids the risk of creating areas with excessively high pH in seawater typical of OAE technologies. The injection of bicarbonates into the sea is done either from the coast through a submarine outfall or from a dedicated ship after rigorous control process to ensure compliance with established injection parameters. This process is based on precise parameter measurements made during the process rather than the use of theoretical models, ensuring a high degree of certainty regarding the amount of CO₂ transformed into bicarbonates.
- **Fast mineralization process of the CO₂:**
 - o The Mixer enhances the dissolution of lime and the CO₂ into calcium bicarbonates in less than 4 minutes.
- **Location flexibility:**

- The location of Limenet plants does not depend on geological storage and it can be installed in any, even remote, coastal location in the world.
- **Scalability:**
 - Considering the availability of limestone⁷, the Limenet process could scale up to multiple Gtons/year of negative emissions.
- **Plants deployment:**
 - Limenet process is innovative but using already existing industrial equipment and known technologies that will allow a quick and consistent scale up.

⁶ Sievert, K., Schmidt, T. S., & Steffen, B. (2024). Considering technology characteristics to project future costs of direct air capture. *Joule*, 8(4), 979–999. <https://doi.org/10.1016/j.joule.2024.02.005>

⁷ Caserini et al., 2022. The Availability of Limestone and Other Raw Materials for Ocean Alkalinity Enhancement. *Global Biogeochemical Cycles*, 36, 5, e2021GB007246. <https://doi.org/10.1029/2021GB007246>

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?⁸ What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Location and Scale:

Limenet is building a complete industrial plant able to produce 1.5k tons of CO₂-free lime per year, which can be used to do CDR. The name of the project is Phoenix. This will be done in Augusta (SR, Sicily, Italy). The facility will be a FOAK demonstrating a 24-hour operation.



Figure 4 Phoenix project

Augusta was chosen for the local network of one of the investors. Even though it is not the best place for renewable electricity, it's close to the limestone quarry of one of Limenet's investors.

Another key motivation for installing the FOAK plant in Augusta port area is that Limenet has already installed its TRL7 Mixer.

⁸ We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

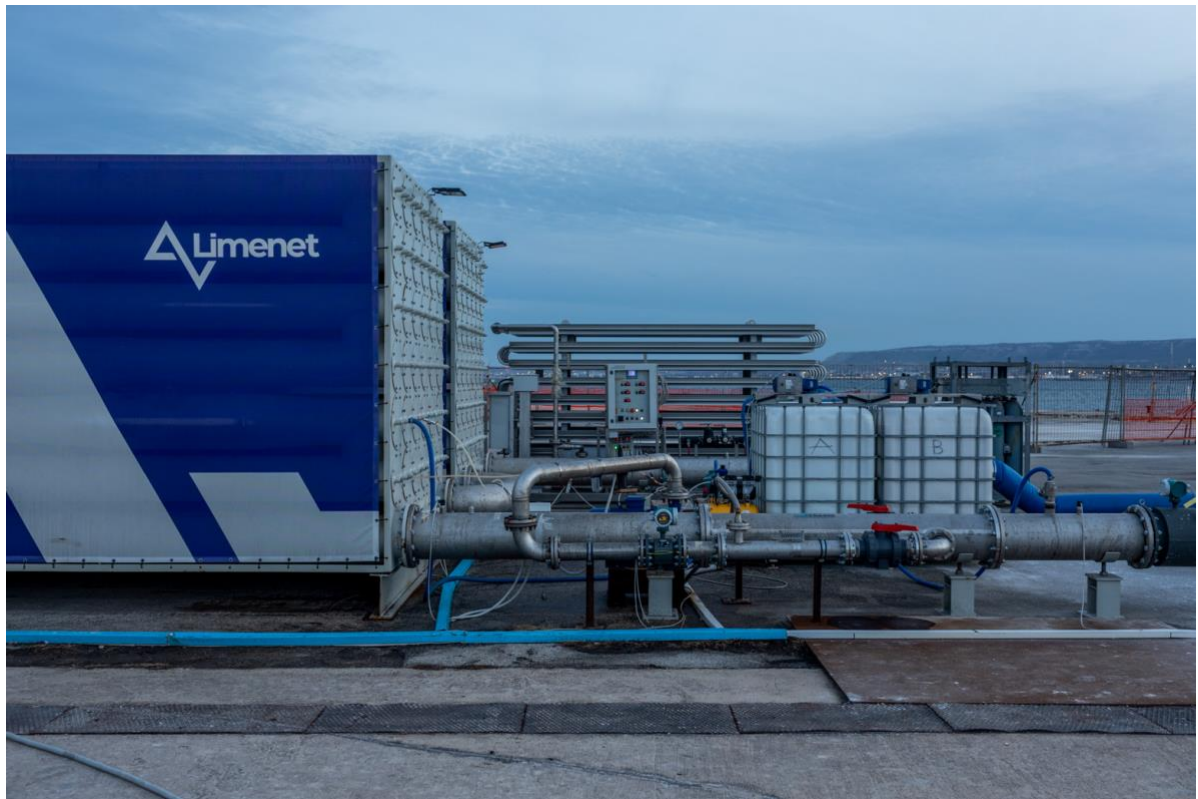


Figure 5 – Limenet Storage Technology, TRL 7 Mixer in Augusta (Sicily, Italy)

At this site, we already have a collaboration agreement with the Port Authority of Eastern Sicily Sea and the Municipality of Augusta. Currently, we hold an occupancy permit for an area of the Augusta commercial port lasting 4 years.

What Limenet is building:

Following Limenet industrial plan, the construction of the Phoenix project consists of 3 sections:

- **Limenet-Calcliner-technology (with an Electric Indirect Kiln and a slaker, under detailed engineering, expected to be erected by early 2026):** starting from limestone, one of the main materials that make up the Earth's crust, this material is fed into the calciner, where it is transformed into calcium oxide and a stream of fossil CO₂, coming from calcination. Into the slaker, the calcium oxide is then hydrated and converted into calcium hydroxide.
- **Limenet-Storage-Technology (with a mixer, already done in 2024):** a dedicated reactor combines seawater with fossil CO₂ from calcination, and 50% of the lime produced in the calciner. The reaction produces a transparent aqueous solution of calcium bicarbonates, whose pH is balanced to match that of seawater. Once this bicarbonate solution is injected into the ocean, it serves as a stable form of CO₂ storage and also helps counteract ocean acidification through its buffering capacity. The mixer module can be installed onshore or placed on a barge for large-scale plants.
- **CDR with the CO₂-free lime:** the other 50% of the lime produced could be considered CO₂-free (NET of the inefficiency and LCA). This lime is used to test 3 different approaches of CDR:
 - Storage of biogenic carbon
 - Mineralization of lime
 - Ocean liming

The reason to test three approaches is to showcase the flexibility of the CO₂ free lime applications.

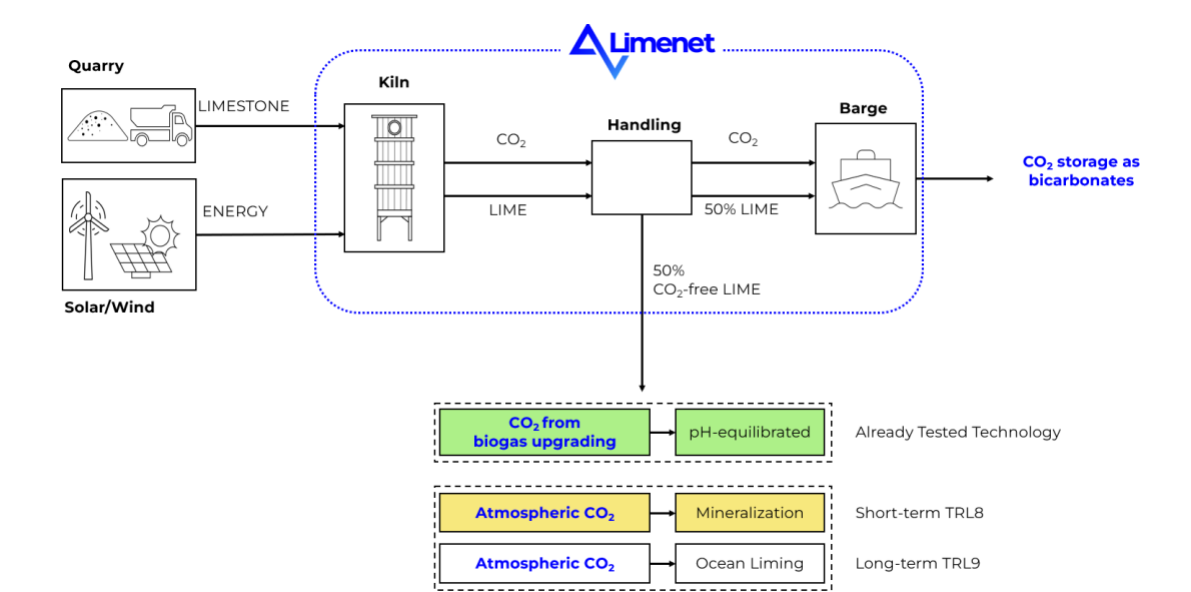


Figure 6 – Phoenix plant schematic and next steps

However, for Frontier, Limenet proposes to use lime in two different ways. To mineralize it and to use it for Ocean liming. This will help us to finish our industrial plan showcase.

The final approach for Limenet, as described in our vision is to make CO₂-free lime to use it for Ocean Liming.

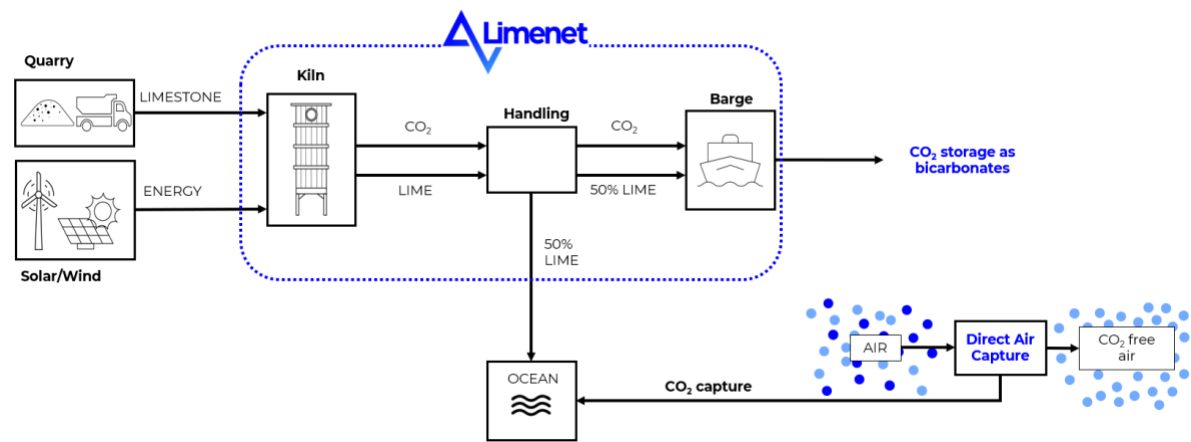


Figure 7 - Limenet TRL9 plant schematic

Current cost breakdown:

The current cost breakdown of the storage plant (mixer) in Augusta (Italy) using biogenic carbon (first approach) is as follows CAPEX 14% OPEX 65%, Personnel 21%, MRV 1%. At the current stage, we are

purchasing both calcium hydroxide and biogenic CO₂ from suppliers to enable the formation of calcium bicarbonates.

Limenet's goal was to test the first way how to use lime for CDR using biogenic carbon. Now the goal is to develop a fully integrated solution in order to produce CO₂-free lime and uses for the other two test: mineralization and Ocean liming.

In order to reduce the cost, in the short to medium term, Limenet has the plan to build a calcination module during 2026 to produce calcium hydroxide, which will complement the existing facility in Augusta. This will allow us to reduce the cost associated with negative emission and LCA impact. Secondly, Limenet aims to eliminate all costs related to the purchase of biogenic CO₂ from external suppliers. The CO₂ will be directly captured on-site at the Limenet facility through mineralization or Ocean liming.

Additionally, as Limenet scales its technology, it plans to sign Power Purchase Agreements (PPAs) to secure a stable supply of renewable electricity that will reduce LCA with a fixed purchasing price. In addition, another step that Limenet aims to take to further reduce its costs is adopting the liming solution for CO₂ removal.

To approach Frontier's cost and scale criteria:

- **Modular approach:** Limenet should scale up the size of the Limenet Calciner technology to enjoy the economies of scale and to reduce the specific cost of the organization and of the services. Limenet would like to keep a modular approach of the Limenet Calciner technology to industrialize each component (calciner, slaker, etc.) that optimizes the manufacturing and the logistic of the equipment to the site.
- **Size and organization optimization**
The size of the Limenet plant that has the best economy of scales is a plant able to produce more than 150 ktons/year of CO₂-free lime. Limenet prefers to deploy the technology in different places starting with small modular plants on a rolling basis, avoiding gigantic projects, long to deliver and difficult to finance.
- **Right location choice where to install the plant:** low LCOE (levelized cost of energy).
- **Increased demand for high quality carbon credits:** enough demand for high quality carbon credit needs to grow to make the Limenet projects "bankable." The other challenge is to have enough initial demand to make the plant running at least at breakeven. This will help Limenet to make operational hours on its new technology, helping to collect operational data, to grow experience and awareness of the problems and make Limenet more ready for the scaling up.
- **Closure of prepurchase agreement:** The signing of pre-purchase agreements would provide strong validation of demand of this new technology approach, demonstrating clear market traction and commercial interest. This kind of endorsement is crucial, not only does it de-risk the project in the eyes of potential investors, but it also significantly strengthens our position when applying for both public and private funding. Indeed, we believe this could facilitate the closing of our seed investment round and support future Series A.

Quantification of Carbon Removal:

The quantification of the carbon removal is done by developing an MRV comprehending two main stages.

Quantification of the production of CO₂-free lime:

The amount of material is measured with the mass of CO₂-free lime that is produced in the plant.

CDR measurement:

Depending on the approach, for each CDR pathway Limenet will produce a PDD according to Limenet ISO 14064-2 certified methodology and PDD (for more details see chapter 5) and the Isometric standard.

There will be three PDD, one for each storage method:

1. Biogenic CO₂ storage, based on the pH-equilibrated certified methodology and PDD;
2. Mineralization of lime, based on the “DAC” and “Carbonated Material Storage and Monitoring” (under Active consultation) Isometric protocols ^{9,10}
3. Ocean Liming, based on the “Ocean Alkalinity Enhancement from Coastal Outfalls” Isometric protocol ¹¹ and biogeochemical models.

In general all processes follows and this formula:

$$CO_{2,removed} = CO_{2,injected}(1) + CO_{2,liming\ mineralization}(2) + CO_{2,ocean\ liming}(3) - Em. CO_{2,calcination} - Em. CO_{2,not\ reacted} - Em. CO_{2,electricity\ consumption} - Em. CO_{2,trasp.} - Em. CO_{2,fug.} - Em. CO_{2,others}$$

Equation 1 - Total CO₂ Emissions removed net of emissions produced

⁹ “Carbonated Material Storage and Monitoring” https://registry.isometric.com/module/carbonated-material-storage/1.1/ctn_1JQS1345B1S0GK3N

¹⁰ “Direct Air Capture” <https://registry.isometric.com/protocol/direct-air-capture/1.1>

¹¹ “Ocean Alkalinity Enhancement from Coastal Outfalls” <https://registry.isometric.com/protocol/ocean-alkalinity-enhancement/1.0>

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Carbon credits demand:

-Demand: The risk is that insufficient early-stage demand for the FOAK could prevent a new project from becoming "bankable".

→Limenet is working to propose credits to different RFP procurement pathways such as Frontier.

→Limenet is exploring other typologies of business models that could co-finance, for the short-term, the industrialization of technology. Limenet has identified a secondary revenue stream through the sale of its electric calcination kilns.

Project execution risks:

-Industrial scale-up: transitioning from a pilot plant to a commercial facility presents significant challenges. The Phoenix project will serve as a FOAK pre-commercial plant, demonstrating the 24/7 operational capabilities of this new technology.

→To mitigate this risk, Limenet has already tested in advance all the key technologies and modules that will be used in the Phoenix project. Limenet successfully tested the TRL7 mixer in Augusta (Sicily, Italy) in the Q3 of 2024. With this facility, the first 14.8 tons of CO₂ were removed for biogenic sources, as certified by the third-party organization Rina S.p.A. Regarding the calcination process, Limenet has successfully developed and tested, in Q1 2025, one of the modules that will comprise the 1.5k tons/year CO₂-free lime calciner.

-Team growth: the risk is not finding the right talent at the right time.

→Limenet is raising equity funding to structure the future team. Talents are sometimes hard to find, that's why Limenet is followed by specific headhunters.

Permits: the risk will be the lengthening of the timeline for obtaining permits for the Phoenix project.

→Because CDR technologies are highly innovative, current regulations often lack appropriate frameworks, leading to permitting delays. Limenet is mitigating this risk by contacting the relevant institutions in advance to explain to them the process and the mission of Limenet. This is done through the help of local experts and specialized companies (e.g. Montana) and by installing the Phoenix project in an industrial area of the Augusta (Italy) Port where should be "feasible" to get permits for industrial activities. In the same municipality, the TRL7 mixer obtained permits to test but not to operate continuously. The authorities that issue the permits for the Phoenix project are the same as the TRL7 mixer.

Permit challenges increase outside the EU, where regulations differ. To address this, Limenet will leverage its Italian showcase and work with an environmental consultancy to navigate local laws. A preliminary area assessment will ensure no conflict with protected or sensitive zones. In Europe and Italy, Rete Natura 2000 protects habitats and species of European importance, while "National Interest Sites" (SIN) are former industrial or heavily polluted sites under strict remediation and land-use controls. Similarly, in the U.S., areas like National Wildlife Refuges, Marine Protected Areas, and Wilderness Areas are protected zones. The U.S. Environmental Protection Agency manages Superfund sites, which are heavily contaminated locations undergoing remediation, analogous to European and Italian SIN zones. These areas are governed by strict regulatory frameworks to avoid further degradation and to safeguard public health and biodiversity and building a plant in such areas could delay the permits' obtainment.

Scientific risks:

The main risks of Limenet's CO₂ storage as calcium bicarbonates in seawater include abiotic/biotic precipitation, dilution, and ecotoxicology. These uncertainties are minimized through experiments conducted in La Spezia (Italy) by Limenet, Politecnico di Milano, and Bicocca University, using mesocosms filled with alkaline seawater from the Limenet process.

- *Abiotic precipitation:* It is a chemical process that occurs when the $\Omega_{\text{aragonite}}$ in the plume that the pH Equilibrated Alkaline Solution forms with the seawater exceeds the value of 5 for about 5 hours.¹²
 - the experimental results¹³ show that, to avoid the abiotic precipitation with final $\Omega_{\text{aragonite}}$ below 5 within about 5 hours¹⁴.
- *Biotic precipitation:* This process involves marine organisms, particularly corals and other calcifying organisms, in the formation of calcium carbonate (CaCO₃) due to alkaline seawater.
 - the tests done by the Bicocca University on the biotic precipitation of calcium bicarbonates in the mesocosms installed in La Spezia show that the increase of biological activity due to increase of seawater alkalinity is minor and that, with a proper plume dilution, it could be considered negligible. These results are in line with other similar mesocosms studies performed by other universities¹⁵.
- *Ecotoxicology:* tests to prove the absence of toxicity in the effluent of Limenet plant through standardized tests.
 - eco-toxicological tests done by Institute for Marine Biological Resources and Biotechnology of the National Research Council (IRBIM-CNR) on different species including significant ones for fishery industry to demonstrate the absence of toxicity that could be related to trace materials present in the Limenet effluent. In addition, further tests can analyze different alkaline feedstock. Metal contamination risk is low because Limenet ensures that the limestone used has no impurities. This activity is co-funded by Carbon to Sea Initiative.
- *Dilutions:*
 - Limenet in collaboration with Politecnico di Milano and Euro-Mediterranean Center on Climate Change (CMCC) is developing a fluid-dynamic model to analyze the behavior of the bicarbonate-rich plume injected into the sea, including dilution effects and time required for safe dispersion before any potential precipitation.

¹² Moras et al., 2022. Ocean alkalinity enhancement – avoiding runaway CaCO₃ precipitation during quick and hydrated lime dissolution, Biogeosciences, 19, 3537–3557. <https://doi.org/10.5194/bg-19-3537-2022>

¹³ Varliero et al. (2024). Assessing the Limit of CO₂ Storage in Seawater as Bicarbonate-Enriched Solutions. Molecules, 29(17), 4069. <https://doi.org/10.3390/molecules29174069>

¹⁴ Moras et al., 2022. Ocean alkalinity enhancement – avoiding runaway CaCO₃ precipitation during quick and hydrated lime dissolution, Biogeosciences, 19, 3537–3557. <https://doi.org/10.5194/bg-19-3537-2022>

¹⁵ Riebesell et al., 2023. Mesocosm experiments in ocean alkalinity enhancement research. Guide to Best Practices in Ocean Alkalinity Enhancement Research, Copernicus Publications, State Planet, 2-oae2023, 6. <https://doi.org/10.5194/sp-2-oae2023-6-2023>

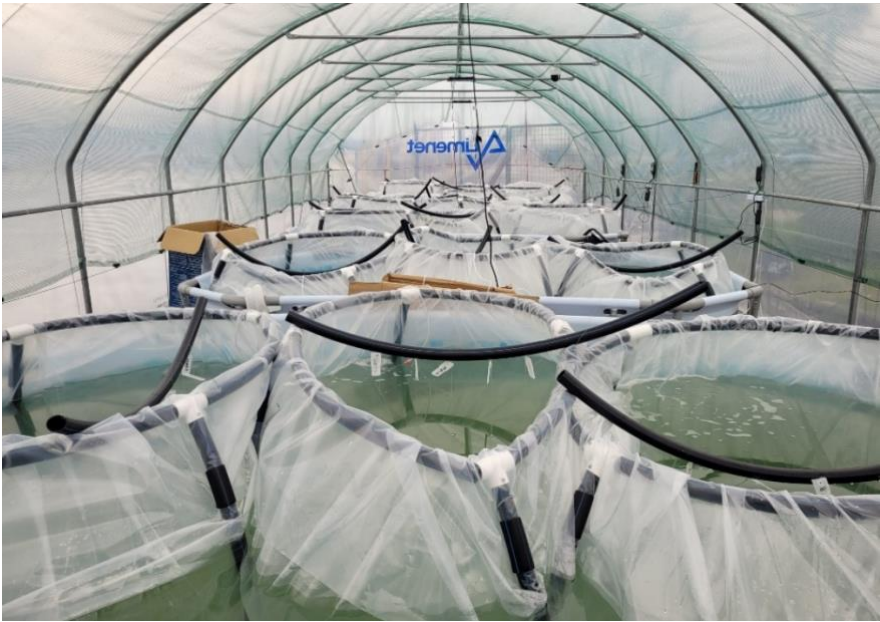


Figure 8 - Mesocosm experiment for assessing abiotic/biotic precipitation and biotic impact on different concentrations of pH-equilibrated-calcium carbonate

Social acceptance risks:

Social acceptance is crucial for CDR but faces challenges such as perceptions of moral hazard, environmental concerns, and local opposition. To mitigate this, Limenet will locate the Phoenix project in Augusta, where the TRL7 unit is already installed and well received. The community is familiar with the project thanks to early engagement, public events, and climate education initiatives.

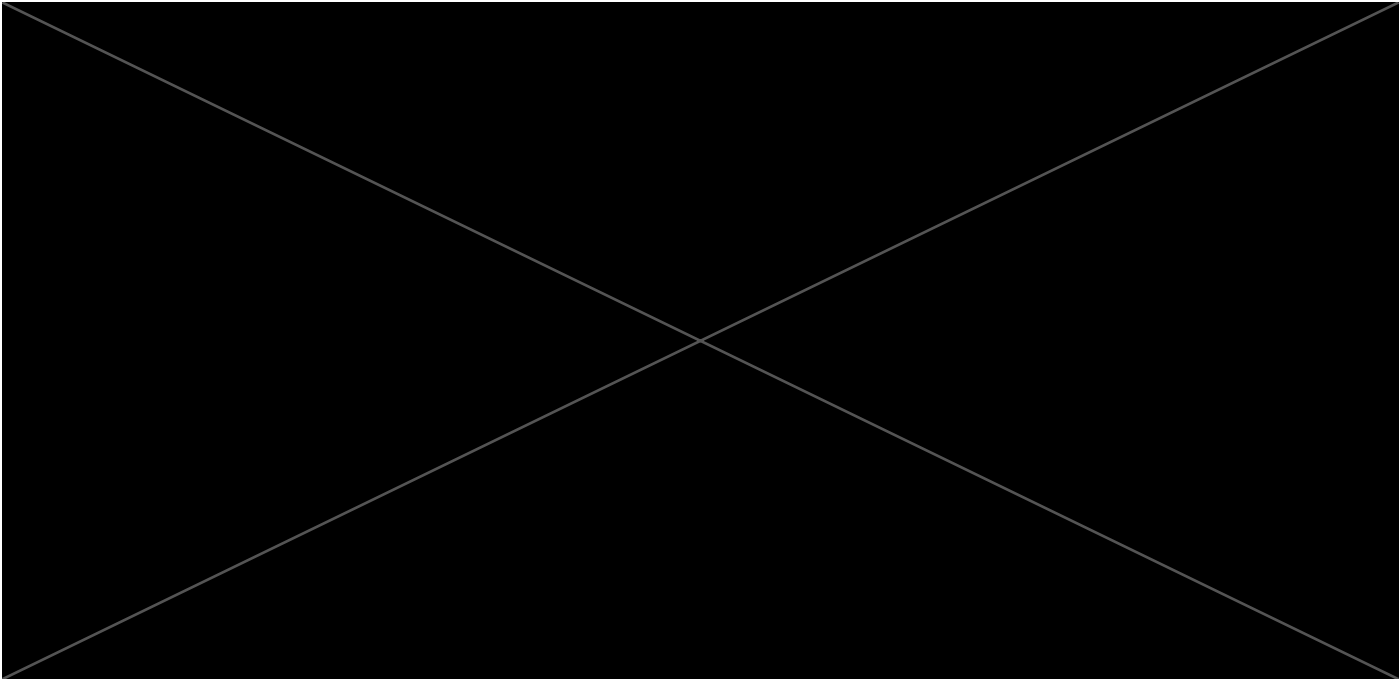
Phoenix Actions Planned

- External Communication: Promote CDR's role in climate action via outreach and public science events to counter moral hazard concerns.
- Community Engagement: Build trust through transparent dialogue and long-term outreach.
 - Education: School collaborations to raise awareness and foster future acceptance; plant visits included.
 - Public Events & Surveys: Share project updates, collect feedback, and highlight local benefits (e.g. ocean-health, jobs).
 - Risk Management: Identify and mitigate possible community impacts (e.g. land use, traffic, air quality) to sustain public support.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	330 tons
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Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	2027: 250 tons (mineralization) 2028: 80 tons (ocean liming)
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	1,816 \$/ton CO ₂
Levelized price (\$/ton CO ₂) ¹⁶ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	1,455 \$/ton CO ₂ (direct costs with no CapEx, according to TEA analysis + 30% indirect cost + 20% net margin)



¹⁶ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).

